Lab 13: How Populations Evolve (Module 13)

BIOL-1

Name: ______________________	**Date:** ______________________

Learning Objectives

By the end of this lab, you should be able to:

1. Calculate allele frequency from genotype counts in a diploid population.
2. Explain how a **bottleneck** can change allele frequencies **without** selection.
3. Classify **directional / stabilizing / disruptive** selection from a before-and-after phenotype distribution.
4. Use Hardy–Weinberg expectations as a **null model** ("what if nothing evolutionary happened?").

Overview

Today you run a **paper simulation** of one imaginary beetle population across **three generations of story time**. You will track **allele frequencies**, watch **random catastrophe** reshuffle them (drift), then watch **non-random survival** (selection) change the distribution. The point is not algebra tricks—it is to feel **when chance matters** and **when the environment steers evolution**.

Materials Needed

- Pencil or pen
- This worksheet
- Calculator (optional)

Warm-up

W1. In one sentence, what is a **population** in biology (the evolutionary meaning—not just "many animals in one place")?

W2. Gene vs alleles: This lab follows one gene with two versions, **R** and **r**. In one sentence, what does it mean to say **R** and **r** are two **alleles**?

W3. Allele frequency: In one sentence, what does **allele frequency** measure—you count copies of an allele, whole individuals, or something else?

W4. Diploid count: One beetle has genotype **Rr**. How many **r** alleles does that one beetle carry? (enter a number)

Part 1: Lakeview Beetles — Baseline (Generation 0)

Gene R controls spot brightness: RR = bright, Rr = mottled, rr = dull. Population is at Hardy–Weinberg expectations for this generation.

In **Generation 0**, scientists genotype **200** beetles:

Genotype	Number of individuals
RR	50
Rr	100
rr	50

1. Count alleles (show work in the margin of your paper if needed).

- Total **R** alleles in the population:

- Total **r** alleles:
- Total alleles (diploid population):

2. Allele frequencies:

- Frequency of **R** (call it p):
- Frequency of **r** (q):
- Check: does $p + q = 1$?

3. Null expectation: If mating is random and nothing evolutionary happens, predict approximate genotype frequencies using p^2, $2pq$, and q^2 (you may round to two decimals).

- Expected **RR**: **Rr**: **rr**:

4. Quick check: Do the observed counts (50 / 100 / 50) match your expectations **exactly**? **Yes / No.** (Real populations often deviate—your job is to notice.)

Part 2: The Dam Break — Bottleneck (Generation 1)

A flash flood strands beetles on a tiny mud island. Only 20 beetles survive—randomly with respect to genotype (no selection yet). Survivors:

Genotype among survivors	Count
RR	4
Rr	8
rr	8

5. Prediction first: Before you calculate, did you expect p to stay exactly **0.5**? **Yes / No.** One sentence why drift should or shouldn't matter here:

6. Calculate new allele frequencies among survivors only **(20 individuals = 40 alleles).**

- Total **R** alleles:
- Total **r** alleles:
- New *p*: New *q*:

7. Interpret: Did the allele frequency change **because birds preferred bright beetles**? **Yes / No.** What **mechanism** caused the change in this part of the story?

Part 3: Bird Arrival — Selection (Generation 2)

Birds colonize the mud island. The visual background is patchy: sometimes bright rock, sometimes deep shadow. Mottled beetles are caught most often because they match neither extreme. Bright and dull extremes both survive better than the middle. After one round of reproduction, the adult phenotype distribution has thinned at the center.

8. Classify the mode of natural selection acting on spot brightness.
Circle one: **Directional / Stabilizing / Disruptive** — defend in two sentences (what happens to the **middle** of the distribution vs the **extremes**?).

9. Mechanism check: Which of the five microevolutionary mechanisms **dominates** this bird episode? (mutation / gene flow / drift / non-random mating / natural selection)

10. Could gene flow **later undo this local adaptation?** Describe a realistic scenario in 2–3 sentences (e.g., bright beetles fly in from the mainland).

Part 4: Hardy–Weinberg as a Foil

Hardy–Weinberg is the "boring universe" where allele frequencies never change. Real populations are more interesting.

11. Name two assumptions of Hardy–Weinberg equilibrium (any two standard ones):

1.
2.

12. In this simulation, which assumptions were clearly violated in Part 2? In Part 3? (short phrases)

- Part 2:
- Part 3:

13. Independent practice: A gene has dominant allele frequency $p = 0.2$. Find q, then **aa** frequency, then **Aa** heterozygote frequency.

- q =
- Freq(**aa**) = q^2 =
- Freq(**Aa**) = $2pq$ =

14. Big picture (3 sentences): Why is it useful to compare **real** genotype frequencies to **Hardy–Weinberg expectations** instead of memorizing "equilibrium" as magic?